120 Days of Quantitative Finance

Month 3: Machine Learning for Finance

Week 10: Advanced Machine Learning Concepts

Day 64: Maximum Likelihood Estimation

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1 Introduction

Maximum Likelihood Estimation (MLE) is one of the most fundamental methods for parameter estimation in statistics and machine learning. It provides a principled way to estimate the parameters of a probabilistic model given observed data.

In quantitative finance, MLE is widely used in:

- Estimating return distributions
- Parameter estimation in volatility models
- Logistic regression for classification
- Hidden Markov Models for regime detection

The central idea behind MLE is simple: choose the parameters that make the observed data most probable.

2 Likelihood Function

Suppose we observe a dataset:

$$X = \{x_1, x_2, ..., x_n\}$$

We assume that the data are generated from a probability distribution with parameter θ.

The probability density function is:

$$p(x|\theta)$$

The **likelihood function** is defined as the joint probability of observing the data given the parameter:

$$L(\theta) = P(X|\theta)$$

Assuming independent observations:

$$L(\theta) = \prod_{i=1}^{n} p(x_i|\theta)$$

The goal of maximum likelihood estimation is to find:

$$\hat{\theta} = \arg\max_{\theta} L(\theta)$$

3 Log-Likelihood

Working directly with the likelihood product can be inconvenient. Therefore we often maximize the **log-likelihood**.

$$\ell(\theta) = \log L(\theta)$$

Using properties of logarithms:

$$\ell(\theta) = \sum_{i=1}^{n} \log p(x_i|\theta)$$

Maximizing the log-likelihood is equivalent to maximizing the likelihood itself.

4 Example: Estimating Mean of Normal Distribution

Suppose returns follow a normal distribution:

$$x_i \sim N(\mu, \sigma^2)$$

The probability density function is:

$$p(x_i|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

The likelihood function becomes:

$$L(\mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

Taking logarithms:

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2$$

Maximizing this with respect to μ gives:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

Thus, the MLE estimator for the mean is simply the sample average.

5 Connection to Machine Learning

Many machine learning models can be interpreted as maximum likelihood estimators.

For example:

Linear Regression

Assume the model:

$$y = X\beta + \epsilon$$

where:

$$\epsilon \sim N(0, \sigma^2)$$

Maximizing the likelihood of this model leads to minimizing:

$$\sum (y_i - X_i\beta)^2$$

which is exactly the **mean squared error loss**.

Therefore, ordinary least squares regression can be interpreted as a maximum likelihood estimator.

6 MLE in Finance

Maximum likelihood estimation is widely used in financial modeling.

Examples include:

- Estimating drift and volatility of asset returns
- Parameter estimation in GARCH models
- Calibration of stochastic processes
- Hidden Markov models for market regimes

For example, if daily returns follow:

$$r_t \sim N(\mu, \sigma^2)$$

the MLE estimates are:

$$\hat{\mu} = \frac{1}{n} \sum r_t$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum (r_t - \hat{\mu})^2$$

These estimates form the basis of many financial risk models.

7 Key Takeaways

- Maximum Likelihood Estimation finds parameters that maximize the probability of observed data.
- The likelihood function measures how probable the data is under a given model.
- Log-likelihood simplifies optimization.
- Many machine learning algorithms are equivalent to maximum likelihood estimators.

- MLE plays a central role in financial modeling and risk estimation.